

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-ab4579ae8c8db4480.jsonl`  
- SHA-256: `a115f5a67d772527bc4f4a9c440bfeac91744f2bdf4a38ea7a79205808a01175`  
- Source modified: `2026-06-06T17:30:30+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:28:43.709Z)

Web Searcher: recent/applied

Research question: "Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ? especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?
5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio)."

Your angle: **recent/applied** ? Surfaces recent (2024-2025) applied/edge work for real-time on-device racket-sport segmentation across sibling sports, catching newer permissive models that fit a single consumer GPU.
Search query: `racket sports rally segmentation 2024 2025 lightweight real-time on device model GPU pickleball table tennis`

Task

Use WebSearch with the query above (or a refined version). Return the top 4-6 most relevant results.

Rank by relevance to the ORIGINAL question, not just the search query. Skip obvious SEO spam/content farms.

Include a short snippet capturing why each result is relevant.

Structured output only.

2. assistant (2026-06-06T17:28:48.174Z)

I'll research this. Let me start with the targeted search query, then run a couple of refined searches in parallel.

3. assistant (2026-06-06T17:28:51.121Z)

Now let me run the searches in parallel.

4. user (2026-06-06T17:29:01.802Z)

Web search results for query: "racket sports rally segmentation 2024 2025 lightweight real-time on device model GPU pickleball table tennis"

Links: [{"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "url": "https://arxiv.org/pdf/2603.13397"}, {"title": "High Speed and Tiny Objects Tracking System in Racquet Sports Videos Using Deep Learning with Trajectory Rectification Feature - ScienceDirect", "url": "https://www.sciencedirect.com/science/article/pii/S1877050925016709"}, {"title": "RacketVision: A Multiple Racket Sports Benchmark for Unified Ball and Racket Analysis | Request PDF", "url": "https://www.researchgate.net/publication/397895011_RacketVision_A_Multiple_Racket_Sports_Benchmark_for_Unified_Ball_and_Racket_Analysis"}, {"title": "RacketVision: A Multiple Racket Sports Benchmark for Unified Ball and Racket Analysis | Proceedings of the AAAI Conference on Artificial Intelligence", "url": "https://ojs.aaai.org/index.php/AAAI/article/view/37362"}, {"title": "RacketVision: A Multiple Racket Sports Benchmark for Unified Ball and Racket Analysis", "url": "https://arxiv.org/html/2511.17045v2"}, {"title": "? Computer Vision for Racket Sports | Tennis & Padel | Paradigma ST", "url": "https://paradigma.dev/cases/computer-vision-subsystems-for-racket-sports-products-tennis-padel/"}, {"title": "GitHub - OrcustD/RacketVision: The official repository of the paper \"RacketVision: A Multiple Racket Sports Benchmark for Unified Ball and Racket Analysis\"", "url": "https://github.com/OrcustD/RacketVision/"}, {"title": "A Deep Learning-Based Framework for Racket Sports Court Registration | Springer Nature Link", "url": "https://link.springer.com/chapter/10.1007/978-3-031-63219-8_2"}]

I found relevant research on racket sports analysis. Here are the key findings from recent work (2024-2025):

Key Projects and Benchmarks

Racket sports, typically represented by badminton, tennis, and table tennis, have garnered widespread global participation and attract research for performance analysis.

RacketVision: A paper accepted by AAAI 2026 (Oral), RacketVision is a comprehensive benchmark covering multiple racket sports. The system uses a two-stage pipeline: RTMDet-M detects racket bounding boxes (3 sport-specific classes), then RTMPose-M estimates 5 keypoints per racket.

Rally Segmentation Approaches

A lightweight, purely audio-based detection model is employed to identify the distinct acoustic signatures of racquet-ball impacts from the broadcast audio track. This approach has been used for efficiently segmenting full match broadcasts into individual rally clips.

Real-Time Performance

TTNet provides real-time temporal and spatial video analysis of table tennis. Additionally, recognition of game events can be achieved with a delay of less than 200 ms in production systems.

Pickleball Integration

While the search results focus heavily on tennis, badminton, and table tennis, the RacketVision framework appears designed to work across multiple racket sports. The modular pipeline approach suggests extensibility to other racket sports like pickleball, though specific pickleball models weren't found in these results.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

5. user (2026-06-06T17:29:03.833Z)

Web search results for query: "badminton rally point temporal segmentation audio shuttle hit detection 2024 deep learning"

Links: [{"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera - PMC", "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC11244353/"}, {"title": "Enhancing Badminton Game Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from Monocular Camera", "url": "https://www.mdpi.com/1424-8220/24/13/4372"}, {"title": "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video", "url": "https://arxiv.org/pdf/2204.01899"}, {"title": "Deep Learning for Sports Video Event Detection: Tasks, Datasets, Methods, and Challenges", "url": "https://arxiv.org/html/2505.03991v3"}, {"title": "A Comparative Analysis of Deep Learning Models and Gradient Computation for Rally Detection in Badminton Videos | SN Computer Science | Springer Nature Link", "url": "https://link.springer.com/article/10.1007/s42979-025-03935-0"}, {"title": "Bridging the Gap: Doubles Badminton Analysis with Singles-Trained Models", "url": "https://arxiv.org/html/2508.13507v1"}, {"title": "Research on badminton video action recognition method based on temporal segmentation network | Proceedings of the 2024 International Conference on Intelligent Education and Computer Technology", "url": "https://dl.acm.org/doi/10.1145/3687311.3687375"}]

Based on the search results, here are the key findings related to badminton rally point temporal segmentation, audio shuttle hit detection, and deep learning approaches from 2024:

Shuttlecock Tracking and Hit Detection

A 2024 study uses TrackNet, a deep neural network designed for tracking small objects, to extract the flight trajectory of the shuttlecock, and the YOLOv7 model to identify whether the player is swinging. The research proposes a shot refinement algorithm to obtain the correct hitting moment, as both TrackNet and YOLOv7 may have detection misses and false detections.

Temporal Segmentation in Badminton

Shot segmentation is critical for identifying the shots of a rally, where a shot happens when a player hits the shuttle with her racket and ends right before the opposing player hits the shuttle or if the shuttle hits the court, making hit event identification equivalent to video segmentation into successive shots.

Deep Learning Methods for Action Recognition

A 2024 approach uses an attention mechanism-temporal segmentation network constructed through badminton action meta-video dataset training for predicting forehand shot, backhand shot, headshot shot, and pick shot actions. The method achieved recall of 91.2% and precision of 91.6%.

Advanced Architectures

Recent work uses Spatial-Temporal Graph Convolutional Networks (ST-GCN) backbone pre-training based on shot and notshot segments for learning effective spatiotemporal representations, as ST-GCN is well-established for strong performance in skeleton-based action recognition.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. user (2026-06-06T17:29:04.188Z)

Web search results for query: "ShuttleSet badminton dataset rally stroke license shuttlecock tracking 2024"

Links: [{"title": "VideoBadminton: A Video Dataset for Badminton Action Recognition | Request PDF", "url": "https://www.researchgate.net/publication/388099044_VideoBadminton_A_Video_Dataset_for_Badminton_Action_Recognition"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://arxiv.org/pdf/2306.04948"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining", "url": "https://dl.acm.org/doi/10.1145/3580305.3599906"}, {"title": "BST: Badminton Stroke-type Transformer for Skeleton-based Action Recognition in Racket Sports", "url": "https://arxiv.org/html/2502.21085v2"}, {"title": "Few-Shot Precise Event Spotting via Unified Multi-Entity Graph and Distillation", "url": "https://arxiv.org/pdf/2511.14186"}, {"title": "[2306.04948] ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis", "url": "https://arxiv.org/abs/2306.04948"}, {"title": "ShuttleSet: A Human-Annotated Stroke-Level Singles Dataset for Badminton Tactical Analysis | Request PDF", "url": "https://www.researchgate.net/publication/372910678_ShuttleSet_A_Human-Annotated_Stroke-Level_Singles_Dataset_for_Badminton_Tactical_Analysis"}, {"title": "(PDF) ShuttleSet22: Benchmarking Stroke Forecasting with Stroke-Level Badminton Dataset", "url": "https://www.researchgate.net/publication/371909519_ShuttleSet22_Benchmarking_Stroke_Forecasting_with_Stroke-Level_Badminton_Dataset"}, {"title": "ShuttleSet22: Benchmarking Stroke Forecasting with Stroke-Level Badminton Dataset", "url": "https://arxiv.org/html/2306.15664v1"}]

Based on the search results, here's what I found about ShuttleSet:

ShuttleSet Overview

ShuttleSet is the largest publicly-available badminton singles dataset with annotated stroke-level records, containing 104 sets, 3,685 rallies, and 36,492 strokes in 44 matches between 2018 and 2021 with 27 top-ranking men's singles and women's singles players.

Annotation Details

ShuttleSet is manually annotated with a computer-aided labeling tool to increase the labeling efficiency and effectiveness of selecting the shot type with a choice of 18 distinct classes, the corresponding hitting locations, and the locations of both players at each stroke.

Technical Features

The dataset includes keypoint information such as players' 2D poses, shuttlecock positions, and court corners, with shuttlecock tracking performed using TrackNetV3.

Related Research

ShuttleSet features 18 types of shots, player positions, and hitting locations, along with benchmarks for stroke influence, forecasting, and movement analysis. The dataset has also been used as a basis for extended research, such as ShuttleSet22, which was collected from high-ranking matches in 2022 and consists of 30,172 strokes in 2,888 rallies in the training set, 1,400 strokes in 450 rallies in the validation set, and 2,040 strokes in 654 rallies in the testing set.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

7. assistant (2026-06-06T17:29:11.043Z)

These are strong hits. Let me dig into the most promising recent/applied ones to verify the audio-segmentation and licensing details, and run one more search focused on real-time/edge action spotting and pose-based state machines.

8. user (2026-06-06T17:29:22.823Z)

Web search results for query: "precise event spotting sports edge real-time T-DEED action spotting lightweight 2024 2025 tennis badminton"

Links: [{"title": "T-DEED: Temporal-Discriminability Enhancer Encoder-Decoder for Precise", "url": "https://openaccess.thecvf.com/content/CVPR2024W/CVsports/papers/Xarles_T-DEED_Temporal-Discriminability_Enhancer_Encoder-Decoder_for_Precise_Event_Spotting_in_Sports_CVPRW_2024_paper.pdf"}, {"title": "Daily Papers - Hugging Face", "url": "https://huggingface.co/papers?q=ball+action+spotting"}, {"title": "An integrated deep learning framework for real-time badminton action recognition and pose estimation with shot-level semantic refinement: Australian Journal of Electrical and Electronics Engineering: Vol 0, No 0 - Get Access", "url": "https://www.tandfonline.com/doi/full/10.1080/1448837X.2025.2553948"}, {"title": "Calendar 2025: World Championships, Cricket World Cups, Hockey Asia Cup and more - ESPN", "url": "https://www.espn.com/olympics/story/_/id/43289570/calendar-2025-schedule-indian-sports-world-championships-cricket-world-cups-hockey-asia-cup-more"}, {"title": "World Table Tennis Calendar", "url": "https://www.worldtabletennis.com/events_calendar"}, {"title": "Wearable sensing for badminton stroke recognition with one-dimensional convolutional neural network | Scientific Reports", "url": "https://www.nature.com/articles/s41598-025-25158-2"}, {"title": "Tournaments | Tournamentsoftware. ...", "url": "https://www.tournamentsoftware.com/find?StatusFilterID=2"}, {"title": "Tournamentsoftware.com", "url": "https://www.tournamentsoftware.com/"}, {"title": "Strategy analysis of badminton players using deep learning from IMU and UWB wearables", "url": "https://ouci.dntb.gov.ua/en/works/7AJkK224/"}, {"title": "Calendar | 2026 TOURNAMENTS - REMAINING", "url": "https://corporate.bwfbadminton.com/events/calendar/"}]

Based on the search results, I found relevant information about T-DEED and action spotting in sports:

T-DEED: Temporal-Discriminability Enhancer Encoder-Decoder

T-DEED is a method for precise event spotting in sports videos that addresses multiple challenges including the need for discriminability among frame representations, high output temporal resolution to maintain prediction precision, and the necessity to capture information at different temporal scales to handle events with varying dynamics.

The architecture features an encoder-decoder for leveraging multiple temporal scales and achieving high output temporal resolution, along with temporal modules designed to increase token discriminability. T-DEED achieves state-of-the-art performance on the FigureSkating and FineDiving datasets.

Action Recognition in Badminton and Tennis

The search results also reveal several related approaches for action spotting in racket sports:

The Badminton Stroke-type Transformer (BST) outperforms previous state-of-the-art on the largest

publicly available badminton video dataset (ShuttleSet), another badminton dataset (BadmintonDB), and a tennis dataset (TenniSet).

Additionally, a comprehensive badminton action analysis framework integrates human pose estimation and temporal tracking based on deep learning, incorporating enhanced YOLOv8 detection networks, pose-guided skeleton fusion modules, and trajectory modelling under real-world badminton match conditions.

These approaches represent recent advances (2024-2025) in lightweight, real-time action spotting for tennis and badminton sports analysis.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. user (2026-06-06T17:29:28.062Z)

RacketVision Benchmark Summary

(1) Rally/Point Temporal Segmentation & Audio-Based Hit Detection

The benchmark does **not** use audio-based hit detection. Instead, it employs a simpler approach: "crowd-sourced annotators segments these raw videos into valid clips. A clip is defined as a continuous segment of 5-10 seconds where the ball is actively in play." The segmentation is manual and visual, not automated through audio analysis.

(2) Models, Computational Requirements & Real-Time Capability

The benchmark evaluates several model architectures:

- **RTMDet** and **RTMPose** (real-time capable)
- **YOLO11** (described as "state-of-the-art vision model for real-time object detection")
- **TrackNetV3** (specialized for tracking small, high-speed objects)
- **LSTM** and **Transformer** backbones for trajectory prediction

The document does **not specify** GPU requirements, memory needs, or inference times. While some models are marketed as "real-time," no computational specifications are provided.

(3) Dataset & Code License

The document provides **no information** regarding licensing terms for the dataset or code release.

(4) Sports Coverage

RacketVision covers exactly three sports: "table tennis, tennis, and badminton." It does not include pickleball.

10. assistant (2026-06-06T17:29:36.216Z)

I have a clear picture now. Let me verify one more high-value lead ? the audio-based rally segmentation mentioned in the first search (it appears to be from TennisExpert) and the ShuttleSet licensing ? to ground the recommendation.

11. user (2026-06-06T17:29:44.679Z)

Web search results for query: "TennisExpert sports video understanding audio impact rally segmentation lightweight model 2025"

Links: [{"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "url": "https://arxiv.org/pdf/2603.13397"}, {"title": "TennisExpert: Towards Expert-Level Analytical Sports Video Understanding", "url": "https://arxiv.org/html/2603.13397v2"}, {"title": "(PDF) TennisExpert:"}]

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Towards Expert-Level Analytical Sports Video Understanding", "url": "https://www.researchgate.net/publication/402480116_TennisExpert_Towards_Expert-Level_Analytical_Sports_Video_Understanding"}, {"title": "(PDF) Tennis video abstraction from audio and visual cues", "url": "https://www.researchgate.net/publication/4146865_Tennis_video_abstraction_from_audio_and_visual_cues"}, {"title": "TennisTV: Do Multimodal Large Language Models Understand Tennis Rallies?", "url": "https://arxiv.org/pdf/2509.15602"}, {"title": "TennisTV: Do Multimodal Large Language Models Understand Tennis Rallies?", "url": "https://arxiv.org/html/2509.15602v1"}]]
```

Based on the search results, here's what I found about TennisExpert and related tennis video understanding research:

TennisExpert Overview

TennisExpert is a comprehensive tennis understanding system with two tightly integrated components. The first is a fine-grained video semantic parser that parses raw video into compact, time-aligned semantic representations in real time, including scoreboard states, shot sequences, ball bounces, and player locations.

Audio-Based Rally Segmentation

A key innovation in TennisExpert is its approach to rally segmentation. The system employs a lightweight, purely audio-based detection model that trains a neural network to identify the distinct acoustic signatures of racquet-ball impacts from the broadcast audio track. The model is made robust against complex ambient stadium noises by augmenting training data with various background sounds like crowd cheering, clapping, and announcer voices, allowing the model to reliably distinguish true hitting events from similar impulsive noises.

TennisVL Dataset

TennisExpert is built on TennisVL, a large-scale tennis benchmark comprising 202 broadcast matches (471.9 hours) segmented into 40,523 rally-level clips, with each clip annotated with high-quality natural language captions with expert-level analysis.

Performance Results

Experiments show that TennisExpert consistently outperforms strong proprietary baselines, including GPT-5, Gemini, and Claude.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. user (2026-06-06T17:29:53.777Z)

RacketVision Licensing and Features

****License:**** RacketVision is released under the "MIT License," which is commercial-friendly and permissive.

****Sport Coverage:**** The dataset covers badminton, table tennis, and tennis?so yes, badminton is included.

****Rally Segmentation:**** The documentation does not explicitly mention rally segmentation as a dedicated feature. However, the data is organized by rally structure (e.g., `<sport>/all/<match>/csv/`).`

****Provided Models and Weights:**** The repository includes pre-trained checkpoints for three core modules:

1. ****BallTrack**** ? TrackNetV3 for ball detection and tracking
2. ****RacketPose**** ? RTMDet-M (detection) and RTMPose-M (5-keypoint pose estimation)

3. ****TrajPred**** ? CrossLSTM-Attn for trajectory prediction

All weights are downloadable via ``python download_checkpoints.py`` from the Hugging Face model repository.

13. assistant (2026-06-06T17:29:57.723Z)

I have everything I need. The TennisExpert paper is the most directly relevant ? it's a recent (2025/2026) applied system whose rally-segmentation component is exactly the audio-based hit detection approach the user is weighing, and it explicitly benchmarks against Gemini/GPT-5/Claude (the VLM-teacher angle). RacketVision is MIT-licensed with badminton coverage and downloadable weights. Let me return the ranked results.

14. user (2026-06-06T17:30:23.710Z)

Structured output provided successfully

15. assistant (2026-06-06T17:30:30.193Z)

Returned 6 ranked results via StructuredOutput. Top finding is TennisExpert (arXiv 2603.13397) ? its lightweight audio-only racquet-impact detector for rally segmentation, plus its local-parser-beats-GPT-5/Gemini/Claude result, directly addresses the audio-vs-VLM-teacher question. RacketVision (AAAI 2026, MIT-licensed, badminton coverage, downloadable real-time weights) and the 2024 Sensors badminton hit-fusion paper (targets the WASB-recall bottleneck) round out the high-relevance tier, with T-DEED (lightweight TAL), ShuttleSet (distillation/eval labels ? licensing caveat), and BST (pose state-machine building block) as complementary leads.